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LEARNER GUIDE

For



Agentic AI for Digital Marketing

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Learning Outcomes

- LO1: Use generative-AI deep research and agentic workflows to formulate a digital marketing strategy, measurable objectives, KPIs and ROI model.
- LO2: Evaluate channel and campaign ROI, synthesise RACE/SOSTAC strategy models, and align recommendations with overarching marketing objectives.
- LO3: Lead the creation of an integrated online presence across web, email, social and emerging channels with human oversight.

How to Use This Guide

Purpose: This guide contains the detailed build procedures intentionally omitted from the visual slide deck. The 15 labs form one connected Northstar Launch campaign automation from research to content, supervised publishing, measurement and optimisation.

Safe mode: All workflows import inactive. Mock Excel workbooks contain synthetic data. Credentialed live-publishing nodes are disabled and point to an invalid example domain. Use dry-run mode unless the trainer authorises a sandbox account.

Human oversight: A human reviewer owns research acceptance, strategy, brand/compliance disposition, the exact publication payload and every budget or optimisation decision.

Evidence standard: For each lab, retain the execution output, acceptance checklist and a short explanation of the failure control. Never include API keys or personal data in evidence.

Environment Setup

1. Create or sign in to a training n8n workspace. Use a dedicated project named TGS-2025056988-Labs.
2. Download the complete labs folder from the LMS and keep every lab folder intact.
3. Confirm each lab contains README.md, workflow.json, a data/ Excel workbook, starter/, solution/ and evidence/.
4. In n8n, choose Import from File and select the lab's workflow.json. Do not activate it.
5. Open the mock workbook and read the READ ME and Data Dictionary sheets before mapping data.
6. Run workflows manually with synthetic data first. Inspect INPUT and OUTPUT for every node before continuing.

7. Keep all live credentials outside the lab folder. Never paste secrets into Code, Set or Sticky Note nodes.

8. After each run, export only synthetic output into the evidence folder and complete the evidence checklist.

Connected Campaign Architecture

Research → Strategy → Content → QA → Human Approval → Social Publishing → Analytics → Optimisation

The campaign ID CMP-NS-001 and supplied artifact IDs preserve lineage. Do not rename them while completing the connected run.

Lab 01 — Campaign Brief Intake & KPI Contract

Criteria	A1, A2
Objective	Translate a business brief into measurable goals and channel KPIs
Output	A validated campaign object with objectives, KPI contracts, owners and guardrails
Mock data	campaign_brief.xlsx
Metric	brief_completeness = populated_required_fields / 8

Input contract

Field	Purpose
campaign_id	Mapped input/output field for Lab 1
objective	Mapped input/output field for Lab 1
audience	Mapped input/output field for Lab 1
budget_sgd	Mapped input/output field for Lab 1
start_date	Mapped input/output field for Lab 1
end_date	Mapped input/output field for Lab 1
owner	Mapped input/output field for Lab 1
risk_tier	Mapped input/output field for Lab 1

Detailed step-by-step procedure

1. Import workflow.json into n8n and keep it inactive.
2. Open campaign_brief.xlsx and review the Campaign Brief sheet.
3. Copy the first row into the Brief Input node's JSON fields.
4. Execute the workflow manually.
5. Inspect Validate Required Fields for missingFields.
6. Confirm Build KPI Contract produces metric, formula, source, target and owner.
7. Set risk_tier to high and run again.
8. Confirm the review branch is selected.
9. Export the execution output to evidence/execution-output.json.
10. Record the acceptance result in evidence/checklist.md.

Verification

Output contains campaign_id CMP-NS-001, at least three KPI contracts, one owner per KPI and a human-review flag.

Failure and control

Risk: Missing owner or budget can cause unbounded spend.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 02 — Evidence Research & Source Scoring

Criteria	K2
Objective	Collect research signals with citation lineage and confidence scoring
Output	A research evidence table with claim, source, recency, reliability and composite score
Mock data	research_sources.xlsx
Metric	evidence_score = relevance × reliability × recency

Input contract

Field	Purpose
source_id	Mapped input/output field for Lab 2
publisher	Mapped input/output field for Lab 2
url	Mapped input/output field for Lab 2
published_date	Mapped input/output field for Lab 2
claim	Mapped input/output field for Lab 2
relevance	Mapped input/output field for Lab 2
reliability	Mapped input/output field for Lab 2
recency	Mapped input/output field for Lab 2

Detailed step-by-step procedure

1. Import workflow.json and open research_sources.xlsx.
2. Inspect the field definitions on the Data Dictionary sheet.
3. Paste the mock rows into Research Inputs.
4. Execute Normalise Evidence and inspect ISO dates and numeric scores.
5. Confirm Score Sources calculates evidence_score.
6. Change reliability of SRC-002 to 0.90 and rerun.
7. Verify Confidence Gate routes records at or above 0.60 to Accepted Evidence.
8. Verify rejected records retain rejection_reason.
9. Save accepted rows to evidence/accepted-research.csv.
10. Document one source-quality limitation.

Verification

Every accepted claim retains source_id, URL, published date and evidence_score; no rejected row reaches synthesis.

Failure and control

Risk: Low-quality sources can contaminate every downstream asset.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 03 — Audience Signal Synthesis & Opportunity Ranking

Criteria	K1, K2
Objective	Synthesize evidence into auditable audience jobs, pains, triggers and opportunities
Output	Ranked audience opportunities linked to evidence IDs
Mock data	audience_signals.xlsx
Metric	ICE = impact × confidence / effort

Input contract

Field	Purpose
signal_id	Mapped input/output field for Lab 3
segment	Mapped input/output field for Lab 3
job	Mapped input/output field for Lab 3
pain	Mapped input/output field for Lab 3
trigger	Mapped input/output field for Lab 3
evidence_ids	Mapped input/output field for Lab 3
impact	Mapped input/output field for Lab 3
confidence	Mapped input/output field for Lab 3
effort	Mapped input/output field for Lab 3

Detailed step-by-step procedure

1. Import workflow.json and open audience_signals.xlsx.
2. Review each signal's evidence_ids.
3. Load the sample row into Audience Signals.
4. Execute Validate Evidence Links.
5. Remove SRC-001 and confirm validation fails.
6. Restore the link and run Compute ICE Score.
7. Confirm the score is 10.0 for the sample.
8. Review the ranked backlog and mark supported=true only after checking evidence.
9. Save the approved backlog.
10. Record a rejected persona statement and why it was rejected.

Verification

Every opportunity has evidence_ids, an ICE score and explicit supported status before strategy synthesis.

Failure and control

Risk: AI-generated personas may encode stereotypes or unsupported claims.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage.

Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 04 — RACE/SOSTAC Strategy Synthesizer

Criteria	K1, K3, A4
Objective	Map approved opportunities into an integrated strategy and control plan
Output	A strategy model connecting situation, objectives, audience, channels, tactics, actions and control
Mock data	strategy_model.xlsx
Metric	$\text{alignment_rate} = \frac{\text{decisions_with_objective_and_owner}}{\text{total_decisions}}$

Input contract

Field	Purpose
stage	Mapped input/output field for Lab 4
decision	Mapped input/output field for Lab 4
evidence_id	Mapped input/output field for Lab 4
owner	Mapped input/output field for Lab 4
kpi	Mapped input/output field for Lab 4
target	Mapped input/output field for Lab 4
review_date	Mapped input/output field for Lab 4

Detailed step-by-step procedure

1. Import workflow.json and review strategy_model.xlsx.
2. Load approved opportunities from Lab 3.
3. Execute Build RACE Map.
4. Verify each RACE stage names an audience action and channel role.
5. Execute Build SOSTAC Control.
6. Inspect owner, KPI, target and review date.
7. Remove an owner and confirm Alignment Gate fails.
8. Restore the owner and set review_status=approved.
9. Export strategy-draft.json.
10. Compare the automation output with the original business objective.

Verification

All strategy decisions trace to evidence, map to RACE, include control fields and pass human strategy review.

Failure and control

Risk: Channel tactics can drift from the business objective.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 05 — Channel Budget & ROI Simulator

Criteria	K4, A3
Objective	Evaluate channel economics and allocate budget using explicit assumptions
Output	A scenario model for CAC, ROAS, contribution margin and incremental ROI
Mock data	channel_economics.xlsx
Metric	$CAC = \text{spend}/\text{customers}$; $ROAS = \text{revenue}/\text{spend}$; $ROI = (\text{revenue} \times \text{margin} - \text{spend})/\text{spend}$

Input contract

Field	Purpose
channel	Mapped input/output field for Lab 5
spend	Mapped input/output field for Lab 5
impressions	Mapped input/output field for Lab 5
clicks	Mapped input/output field for Lab 5
leads	Mapped input/output field for Lab 5
customers	Mapped input/output field for Lab 5
revenue	Mapped input/output field for Lab 5
gross_margin	Mapped input/output field for Lab 5

Detailed step-by-step procedure

1. Import workflow.json and open channel_economics.xlsx.
2. Load the three sample channel rows.
3. Execute Calculate Economics.
4. Manually verify LinkedIn CAC and ROAS.
5. Confirm Volume Guardrail excludes scenarios with fewer than 10 customers.
6. Change Search customers to 6 and rerun.
7. Inspect the exclusion reason.
8. Restore the value and review ranked scenarios.
9. Enter reviewer_decision and approved_budget.
10. Save the approved allocation to evidence/budget-decision.json.

Verification

Calculated metrics match Excel formulas, low-volume scenarios are flagged and no allocation proceeds without reviewer_decision.

Failure and control

Risk: Optimising ROAS alone may cut channels that create assisted demand.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 06 — Campaign Backlog & Experiment Design

Criteria	A2, A4
Objective	Convert strategy into sequenced work and measurable experiments
Output	A prioritised backlog with hypotheses, variants, sample thresholds and stop rules
Mock data	campaign_backlog.xlsx
Metric	$\text{priority} = \text{expected_impact} \times \text{confidence} / \text{effort}$

Input contract

Field	Purpose
item_id	Mapped input/output field for Lab 6
stage	Mapped input/output field for Lab 6
hypothesis	Mapped input/output field for Lab 6
primary_metric	Mapped input/output field for Lab 6
minimum_sample	Mapped input/output field for Lab 6
stop_rule	Mapped input/output field for Lab 6
owner	Mapped input/output field for Lab 6
status	Mapped input/output field for Lab 6

Detailed step-by-step procedure

1. Import workflow.json and open campaign_backlog.xlsx.
2. Load the approved strategy decisions from Lab 4.
3. Run Create Work Items.
4. Inspect hypothesis, metric and owner fields.
5. Set minimum_sample to zero and confirm validation fails.
6. Restore a realistic sample threshold.
7. Add a stop rule and rerun.
8. Review prioritised items with the campaign owner.
9. Approve only items with evidence and capacity.
10. Export approved-backlog.csv.

Verification

Each ready item has one falsifiable hypothesis, primary metric, minimum sample, stop rule and accountable owner.

Failure and control

Risk: Automated testing without stop rules can expose audiences to harmful variants.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 07 — Content Brief & Prompt Contract

Criteria	K3, A4
Objective	Create a canonical content brief with evidence, brand and output constraints
Output	A versioned JSON content contract suitable for deterministic channel generation
Mock data	content_brief.xlsx
Metric	contract_valid = required_fields_present AND evidence_ids_resolved

Input contract

Field	Purpose
brief_id	Mapped input/output field for Lab 7
campaign_id	Mapped input/output field for Lab 7
audience_job	Mapped input/output field for Lab 7
message	Mapped input/output field for Lab 7
approved_claims	Mapped input/output field for Lab 7
evidence_ids	Mapped input/output field for Lab 7
tone	Mapped input/output field for Lab 7
cta	Mapped input/output field for Lab 7
prohibited_terms	Mapped input/output field for Lab 7

Detailed step-by-step procedure

1. Import workflow.json and open content_brief.xlsx.
2. Load EXP-001 and the approved research evidence.
3. Run Assemble Brief.
4. Inspect the JSON contract and version.
5. Delete evidence_ids and confirm Schema Validator fails.
6. Restore evidence_ids and run Claim Gate.
7. Add an unapproved claim and confirm it is rejected.
8. Remove the claim and review the final prompt contract.
9. Approve the exact contract hash.
10. Save brief-contract.json in evidence/.

Verification

The approved contract includes evidence IDs, approved claims, prohibited terms, CTA, version and contract hash.

Failure and control

Risk: Unbounded prompts can invent claims or leak confidential context.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 08 — Multi-Channel Content Factory

Criteria	K5, A5
Objective	Transform one approved brief into channel-specific assets
Output	Website, email, LinkedIn, Instagram and short-video drafts with shared message lineage
Mock data	content_matrix.xlsx
Metric	channel_fit = passed_length AND required_fields AND approved_claims_only

Input contract

Field	Purpose
asset_id	Mapped input/output field for Lab 8
channel	Mapped input/output field for Lab 8
format	Mapped input/output field for Lab 8
max_chars	Mapped input/output field for Lab 8
message_angle	Mapped input/output field for Lab 8
cta	Mapped input/output field for Lab 8
status	Mapped input/output field for Lab 8

Detailed step-by-step procedure

1. Import workflow.json and open content_matrix.xlsx.
2. Load the approved BRF-001 contract.
3. Execute Generate Canonical Copy.
4. Inspect the common message and evidence lineage.
5. Run Split Channels and Apply Channel Rules.
6. Verify LinkedIn, email and website outputs differ in format and CTA placement.
7. Set a channel limit below output length and confirm the asset is flagged.
8. Restore the limit and rerun.
9. Inspect Recombine Assets for one asset per requested channel.
10. Save generated-assets.json.

Verification

One approved brief produces distinct channel assets; every asset retains brief_id, evidence_ids and validation status.

Failure and control

Risk: Copying identical text across channels ignores audience intent and platform constraints.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 09 — Brand, Claim & Compliance QA

Criteria	K5, A5
Objective	Apply deterministic controls before human editorial review
Output	A QA report covering schema, brand, claims, privacy, accessibility and channel limits
Mock data	brand_rules.xlsx
Metric	$qa_pass_rate = \frac{passed_rules}{applicable_rules}$

Input contract

Field	Purpose
rule_id	Mapped input/output field for Lab 9
category	Mapped input/output field for Lab 9
pattern	Mapped input/output field for Lab 9
severity	Mapped input/output field for Lab 9
action	Mapped input/output field for Lab 9
owner	Mapped input/output field for Lab 9

Detailed step-by-step procedure

1. Import workflow.json and open brand_rules.xlsx.
2. Load assets from Lab 8 and the mock rules.
3. Run Deterministic Checks.
4. Confirm the word guaranteed creates a high-severity violation.
5. Remove the prohibited claim and rerun.
6. Insert an email address and confirm privacy review is required.
7. Inspect the Risk Classifier routing.
8. Verify only zero-high-severity assets reach Pass Queue.
9. Assign review owners for medium-risk findings.
10. Export qa-report.json.

Verification

No high-severity asset reaches approval; every medium-risk finding has category, evidence, owner and disposition.

Failure and control

Risk: An AI self-review can miss the same hallucination made during generation.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 10 — Human Approval State Machine

Criteria	A5
Objective	Pause automation until an accountable reviewer approves the exact payload
Output	An approval request, tamper-evident decision record and state transition
Mock data	approval_queue.xlsx
Metric	approval_integrity = decision_hash == current_payload_hash

Input contract

Field	Purpose
approval_id	Mapped input/output field for Lab 10
asset_id	Mapped input/output field for Lab 10
payload_hash	Mapped input/output field for Lab 10
risk	Mapped input/output field for Lab 10
reviewer	Mapped input/output field for Lab 10
status	Mapped input/output field for Lab 10
decision_at	Mapped input/output field for Lab 10
comment	Mapped input/output field for Lab 10

Detailed step-by-step procedure

1. Import workflow.json and open approval_queue.xlsx.
2. Load the QA-passed asset from Lab 9.
3. Execute Create Approval Record and note payload_hash.
4. Open the simulated approval form URL shown by the workflow.
5. Approve the asset with a reviewer comment.
6. Confirm Wait for Decision resumes.
7. Change the payload after approval and rerun Verify Payload Hash.
8. Confirm the workflow escalates the mismatch.
9. Restore the approved payload and rerun.
10. Save approval-record.json with reviewer, timestamp and hash.

Verification

Only APPROVED records with a matching payload hash proceed; rejected and changed payloads cannot publish.

Failure and control

Risk: A reviewer may approve one version while a changed payload is later published.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 11 — Social Publishing Dry-Run & Idempotency

Criteria	K5, A5
Objective	Build safe per-channel publication payloads and prevent duplicates
Output	Validated dry-run payloads for LinkedIn, Instagram and webhook-based channels
Mock data	publishing_queue.xlsx
Metric	duplicate_rate = duplicate_attempts / publish_attempts

Input contract

Field	Purpose
publish_id	Mapped input/output field for Lab 11
asset_id	Mapped input/output field for Lab 11
channel	Mapped input/output field for Lab 11
scheduled_at	Mapped input/output field for Lab 11
idempotency_key	Mapped input/output field for Lab 11
approval_id	Mapped input/output field for Lab 11
dry_run	Mapped input/output field for Lab 11
status	Mapped input/output field for Lab 11

Detailed step-by-step procedure

1. Import workflow.json and keep Live Publisher disabled.
2. Open publishing_queue.xlsx and load PUB-001.
3. Execute Build Channel Payload.
4. Inspect the target account placeholder, media, text and scheduled time.
5. Run Idempotency Store and Already Published?.
6. Execute the dry-run branch and inspect the simulated 202 response.
7. Execute the same publish_id again.
8. Confirm the duplicate branch prevents a second publication.
9. Change channel to instagram and inspect channel-specific validation.
10. Save publication-log.json.

Verification

Dry-run payload is channel-valid, approval-linked and idempotent; the credentialed live publisher remains disabled by default.

Failure and control

Risk: Retries can publish the same content multiple times or to the wrong account.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 12 — End-to-End Campaign Orchestrator

Criteria	A4, A5
Objective	Connect research, strategy, content, QA, approval and publishing workflows
Output	A parent workflow that calls each stage, records state and stops safely on failure
Mock data	orchestrator_runs.xlsx
Metric	stage_success_rate = successful_stages / attempted_stages

Input contract

Field	Purpose
run_id	Mapped input/output field for Lab 12
campaign_id	Mapped input/output field for Lab 12
stage	Mapped input/output field for Lab 12
status	Mapped input/output field for Lab 12
started_at	Mapped input/output field for Lab 12
ended_at	Mapped input/output field for Lab 12
evidence_uri	Mapped input/output field for Lab 12
error_code	Mapped input/output field for Lab 12

Detailed step-by-step procedure

1. Import workflow.json plus the workflows from Labs 2, 4, 8, 9, 10 and 11.
2. Map each Execute Workflow node to the imported child workflow.
3. Enable Wait for Sub-Workflow Completion.
4. Run CMP-NS-001 in dry-run mode.
5. Inspect Run Ledger after every stage.
6. Force QA Sub-workflow to return rejected.
7. Confirm Approval and Publishing do not execute.
8. Restore the valid asset and approve it.
9. Confirm the dry-run publisher executes once.
10. Export the complete run ledger and final state.

Verification

A successful run reaches PUBLISHED_DRY_RUN only after research, strategy, content, QA and approval succeed; failures stop downstream execution.

Failure and control

Risk: A parent agent may continue after a failed or unapproved stage.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 13 — Performance Event Ingestion & Attribution

Criteria	K2, K4, A3
Objective	Normalise channel events into a campaign performance contract
Output	A deduplicated fact table joined by campaign, asset, channel and event time
Mock data	performance_events.xlsx
Metric	data_quality = valid_unique_events / received_events

Input contract

Field	Purpose
event_id	Mapped input/output field for Lab 13
campaign_id	Mapped input/output field for Lab 13
asset_id	Mapped input/output field for Lab 13
channel	Mapped input/output field for Lab 13
event	Mapped input/output field for Lab 13
event_time	Mapped input/output field for Lab 13
value_sgd	Mapped input/output field for Lab 13
cost_sgd	Mapped input/output field for Lab 13
source	Mapped input/output field for Lab 13

Detailed step-by-step procedure

1. Import workflow.json and open performance_events.xlsx.
2. Load the sample events.
3. Execute Normalise Schema.
4. Confirm timestamps use ISO 8601 with offset.
5. Duplicate EVT-001 and run Deduplicate Event IDs.
6. Confirm only one fact remains.
7. Remove campaign_id and confirm the event is quarantined.
8. Restore the ID and run Join Campaign Map.
9. Inspect the attribution label and limitation field.
10. Save performance-facts.csv.

Verification

All facts have unique event_id, valid campaign_id, normalised time and an explicit attribution rule or quarantine reason.

Failure and control

Risk: Inconsistent IDs and time zones can misattribute outcomes.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage.

Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 14 — ROI Dashboard & Anomaly Detection

Criteria	K2, K4, A3
Objective	Calculate channel economics and surface statistically meaningful changes
Output	A channel-period metric table with CAC, ROAS, ROI, confidence and anomaly flags
Mock data	roi_dashboard.xlsx
Metric	ROI = $(\text{revenue} \times \text{gross_margin} - \text{spend}) / \text{spend}$; lift = $(\text{current} - \text{baseline}) / \text{baseline}$

Input contract

Field	Purpose
period	Mapped input/output field for Lab 14
channel	Mapped input/output field for Lab 14
spend	Mapped input/output field for Lab 14
customers	Mapped input/output field for Lab 14
revenue	Mapped input/output field for Lab 14
gross_margin	Mapped input/output field for Lab 14
baseline_cv	Mapped input/output field for Lab 14
current_cv	Mapped input/output field for Lab 14

Detailed step-by-step procedure

1. Import workflow.json and open roi_dashboard.xlsx.
2. Load the weekly samples.
3. Execute Aggregate Channel-Period and Calculate ROI.
4. Manually verify one ROI result in Excel.
5. Run Compare Baseline and inspect relative lift.
6. Set customers to 1 and confirm the confidence guardrail flags low volume.
7. Restore the value and check CPA and complaint-rate guardrails.
8. Review the dashboard dataset.
9. Enter analyst_decision for each anomaly.
10. Export dashboard-dataset.json.

Verification

Dashboard metrics match the formulas; anomalies include volume, baseline, guardrail status and analyst decision.

Failure and control

Risk: Small samples can produce dramatic but unreliable lift.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage.

Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

Lab 15 — Optimisation Feedback Loop & Capstone

Criteria	A3, A4, A5
Objective	Propose evidence-led changes without allowing autonomous budget or publishing mutations
Output	A controlled recommendation, experiment update and approved next-cycle campaign plan
Mock data	optimisation_actions.xlsx
Metric	realised_lift = (new_result-baseline)/baseline

Input contract

Field	Purpose
action_id	Mapped input/output field for Lab 15
finding	Mapped input/output field for Lab 15
evidence	Mapped input/output field for Lab 15
proposed_change	Mapped input/output field for Lab 15
expected_impact	Mapped input/output field for Lab 15
risk	Mapped input/output field for Lab 15
owner	Mapped input/output field for Lab 15
decision	Mapped input/output field for Lab 15
next_review	Mapped input/output field for Lab 15

Detailed step-by-step procedure

1. Import workflow.json and open optimisation_actions.xlsx.
2. Load the reviewed dashboard from Lab 14.
3. Execute Generate Recommendation.
4. Inspect evidence, expected impact and uncertainty.
5. Set proposed budget change to 60% and confirm Policy Guardrails rejects it.
6. Restore the change to 10%.
7. Approve the recommendation with an accountable owner.
8. Confirm Update Backlog creates a new experiment rather than mutating a live campaign.
9. Run the full dry-run chain from Lab 12 with the new backlog item.
10. Save the cycle audit and write a capstone reflection.

Verification

No optimisation changes a live campaign automatically; approved recommendations create a measured next-cycle experiment with full lineage.

Failure and control

Risk: A self-optimising agent can chase short-term metrics and exceed delegated authority.

If the check fails, stop the downstream branch, preserve the failed payload, record the error and owner, correct the contract or policy issue, then rerun from the failed stage. Never bypass approval.

Required evidence

- Workflow remains inactive unless authorised.
- Execution output contains synthetic data only.
- Evidence checklist is complete.
- Learner can explain the decision rule and oversight point.

End-to-End Capstone Acceptance

- Research has source lineage and confidence.
- Strategy maps to objectives, owners, KPIs and review dates.
- Assets retain brief and evidence IDs.
- No unresolved high-severity QA issue.
- Approval matches the exact payload hash.
- Publishing is dry-run or authorised sandbox and idempotent.
- Performance events use valid campaign, asset and event IDs.
- Optimisation creates a reviewed next-cycle experiment.

Troubleshooting

Workflow imports but has a warning: Open the named node and confirm the installed n8n version supports its type. Keep the workflow inactive until the warning is resolved.

A field is undefined: Compare the item JSON with the lab Data Dictionary. Correct the exact field name; n8n expressions are case-sensitive.

A branch is not taken: Open the upstream node's OUTPUT, inspect types as well as values, and test the IF/Code rule with one item.

The Wait node does not resume: Use the test webhook URL during a manual run, keep the execution open, and submit a reviewer decision before the test URL expires.

A duplicate publish appears: Confirm publish_id and idempotency_key are stable across retries and check the publication ledger before the publisher.

Metrics differ from Excel: Confirm the same grain, gross-margin assumption, period and zero-division rule are used in both calculations.

References

- n8n documentation — <https://docs.n8n.io/>
- Course page — <https://www.tertiarycourses.com.sg/wsqa-gentic-ai-for-digital-marketing.html>
- Legacy deck and supplied ebooks in reference/.